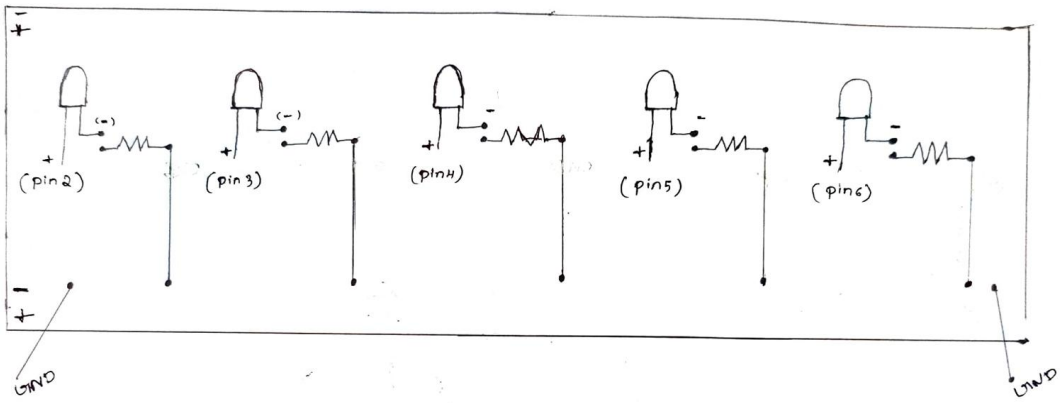
1. Develop a program to blink 5 LEDs back and forth.

```
// LED pin numbers
int ledPins[] = {2, 3, 4, 5, 6};
int numLeds = 5;
void setup() {
  // Set all LED pins as output
  for (int i = 0; i < numLeds; i++) {
    pinMode(ledPins[i], OUTPUT);
  }
}
void loop() {
  // Forward direction (left to right)
  for (int i = 0; i < numLeds; i++) {
    turnOnOnly(i);
    delay(150);
  }
  // Backward direction (right to left)
  for (int i = numLeds - 2; i > 0; i--) { // avoid repeating the first and last LED
    turnOnOnly(i);
    delay(150);
  }
}
// Function to turn on one LED and turn off the rest
void turnOnOnly(int ledIndex) {
  for (int i = 0; i < numLeds; i++) {
    digitalWrite(ledPins[i], i == ledIndex ? HIGH : LOW);
  }
```

}

OUTPUT



2. Develop a program to interface a relay with Arduino board.

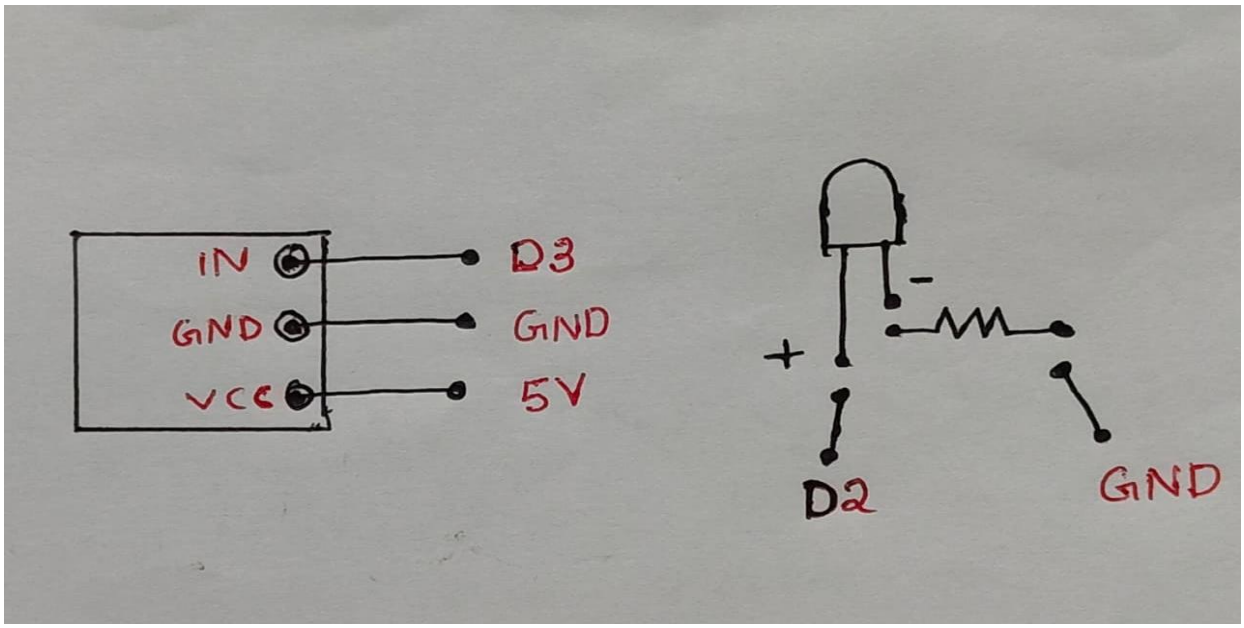
```
const int ledPin = 2;
const int relayPin = 3;

void setup() {
  pinMode(ledPin, OUTPUT);
  pinMode(relayPin, OUTPUT);
}

void loop() {
  // Turn ON LED and Relay
  digitalWrite(ledPin, HIGH);
  digitalWrite(relayPin, LOW);
  delay(1000);

  // Turn OFF LED and Relay
  digitalWrite(ledPin, LOW);
  digitalWrite(relayPin, HIGH);
  delay(1000); // OFF for 1 second
}
```

OUTPUT



3. Develop a program to deploy an intrusion detection system using Ultrasonic and sound sensors.

```
const int trigPin = 9;
const int echoPin = 10;
const int soundPin = 8;
const int buzzer = 7;

long duration;
int distance;

void setup() {
  pinMode(trigPin, OUTPUT);
  pinMode(echoPin, INPUT);
  pinMode(soundPin, INPUT);
  pinMode(buzzer, OUTPUT);
  Serial.begin(9600);
}

void loop() {
  // Ultrasonic sensor


  digitalWrite(trigPin, LOW);
  delayMicroseconds(2);

  digitalWrite(trigPin, HIGH);
  delayMicroseconds(10);
  digitalWrite(trigPin, LOW);
  duration = pulseIn(echoPin, HIGH);
```

```
  // Calculate distance in cm
  distance = duration * 0.034 / 2;

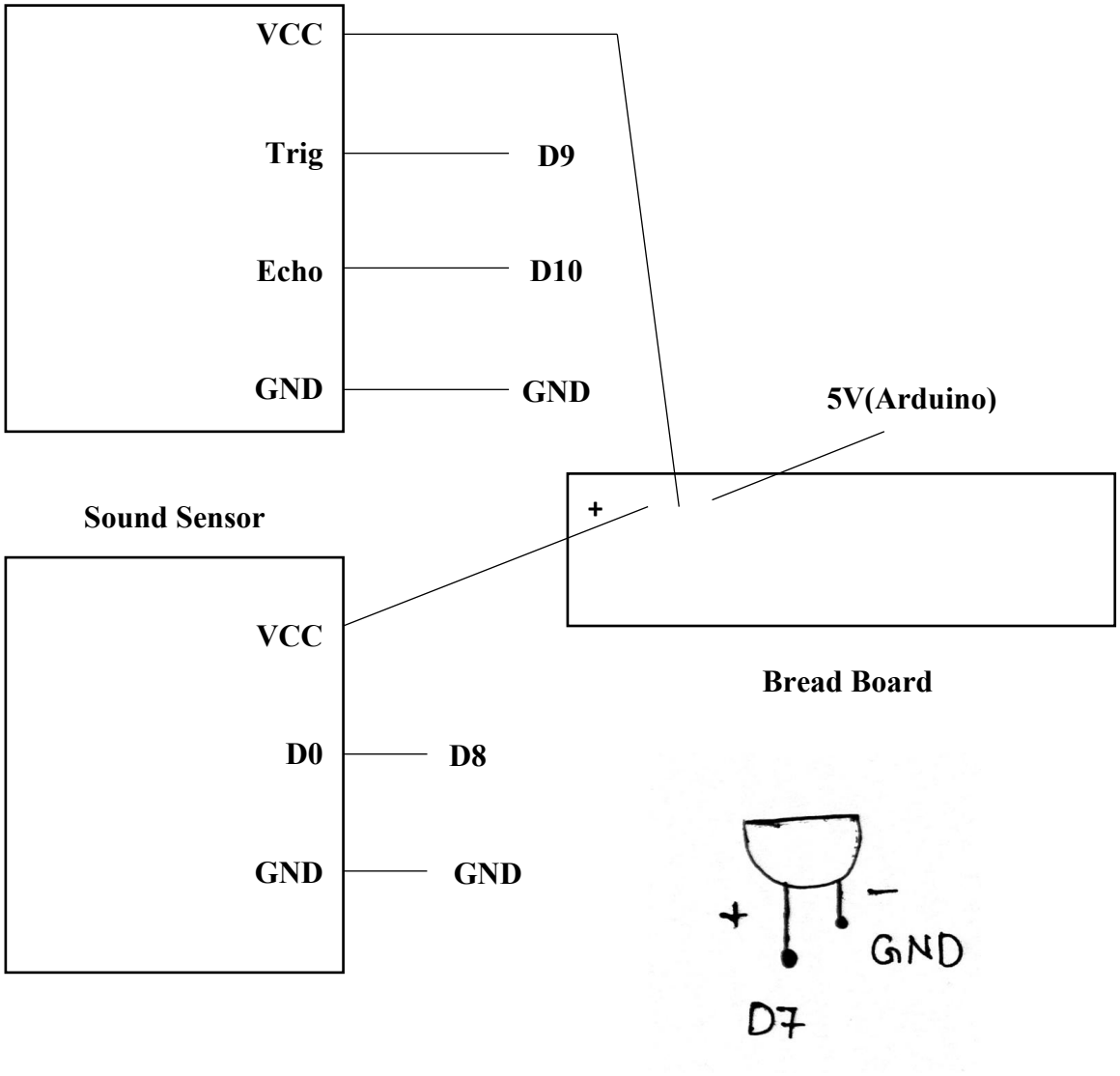
  // Sound sensor
   int soundState = digitalRead(soundPin);
  Serial.print("Distance: ");
  Serial.print(distance);
  Serial.print(" cm");

  if (soundState == LOW) {
    Serial.println(" | Loud sound detected");
  } else {
    Serial.println(" | No loud sound");
  }


  // Intrusion detection
  if (distance < 20 || soundState == LOW) {
    digitalWrite(buzzer, HIGH);
  }
  else {
    digitalWrite(buzzer, LOW);
  }

  delay(500);
}
```

OUTPUT


VCC
Trig
D9
Echo
D10
GND
GND
5V(Arduino)
+
Sound Sensor
VCC
Bread Board
D0
D8
GND
GND
+
−
GND
D7

4. Develop a program to control a DC motor with Arduino board.

```
// Motor control pins
const int IN1 = 8;
const int IN2 = 9;
const int ENA = 10;  // Must be PWM-capable

void setup() {
  pinMode(IN1, OUTPUT);
  pinMode(IN2, OUTPUT);
  pinMode(ENA, OUTPUT);
}

void loop() {
  // Spin motor forward
  digitalWrite(IN1, HIGH);
  digitalWrite(IN2, LOW);

  // Increase speed gradually
  for (int speed = 0; speed <= 255; speed += 5) {
    analogWrite(ENA, speed);
    delay(20);
  }

  delay(1000);  // Run at full speed for a while

  // Decrease speed
  for (int speed = 255; speed >= 0; speed -= 5) {
    analogWrite(ENA, speed);
    delay(20);
```

```
  }

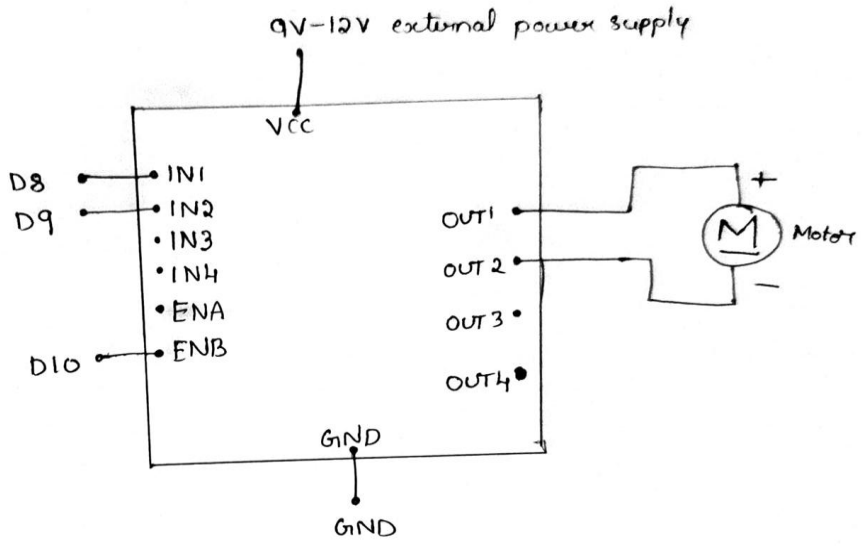
  delay(1000);  // Pause

  // Spin motor in reverse
  digitalWrite(IN1, LOW);
  digitalWrite(IN2, HIGH);

  // Full speed reverse
  analogWrite(ENA, 200);
  delay(2000);

  // Stop motor
  analogWrite(ENA, 0);
  delay(2000);
}
```

OUTPUT



5. Develop a program to deploy smart street light system using LDR sensor.

```
#define LDR_PIN 2      // Digital output from LDR module
#define LED_PIN 13     // LED as streetlight

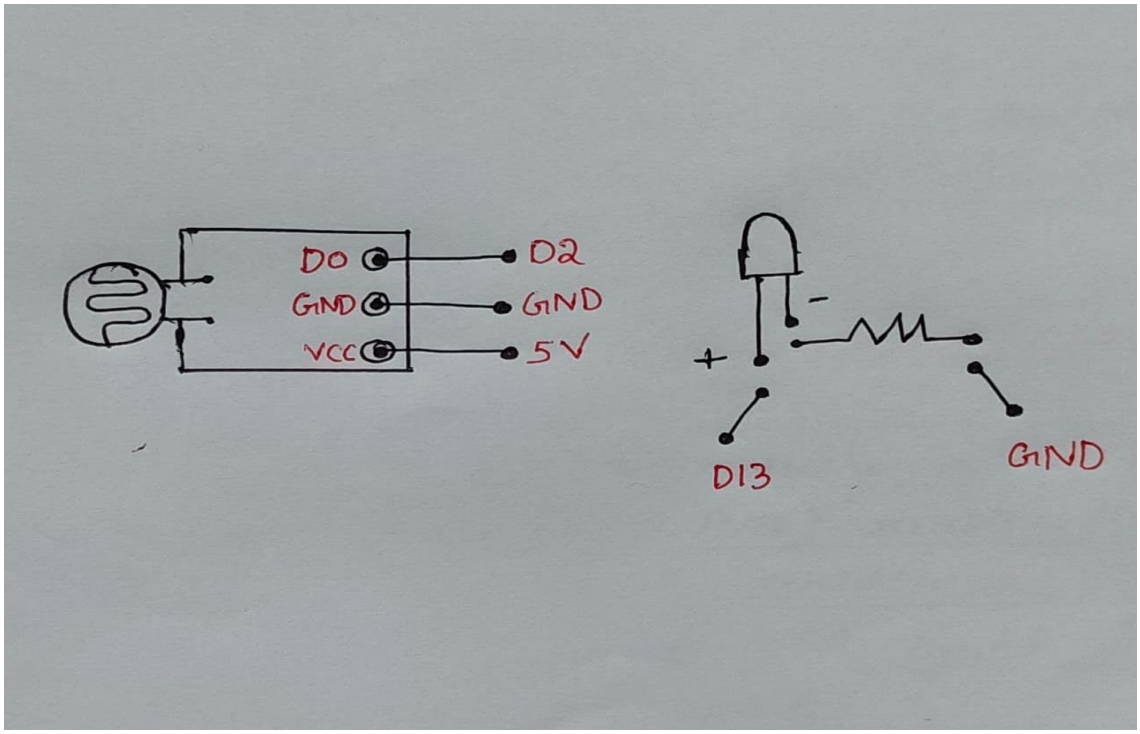
void setup() {
  pinMode(LDR_PIN, INPUT);
  pinMode(LED_PIN, OUTPUT);
}

void loop() {
  int ldrState = digitalRead(LDR_PIN);

  if (ldrState == LOW) {
    // It's dark
    digitalWrite(LED_PIN, LOW);  // Turn ON street light
  } else {
    // It's bright
    digitalWrite(LED_PIN, HIGH);   // Turn OFF street light
  }

  delay(100);  // Small delay for stability
}
```

OUTPUT



6. Develop a program to classify dry and wet waste with the Moisture sensor (DHT22).

```
const int sensorPin = A0;    // Analog input pin for moisture sensor
const int ledPin = 8;        // LED to indicate wet waste
const int buzzerPin = 11;     // Buzzer for wet waste alert

int sensorValue = 0;
int threshold = 500;

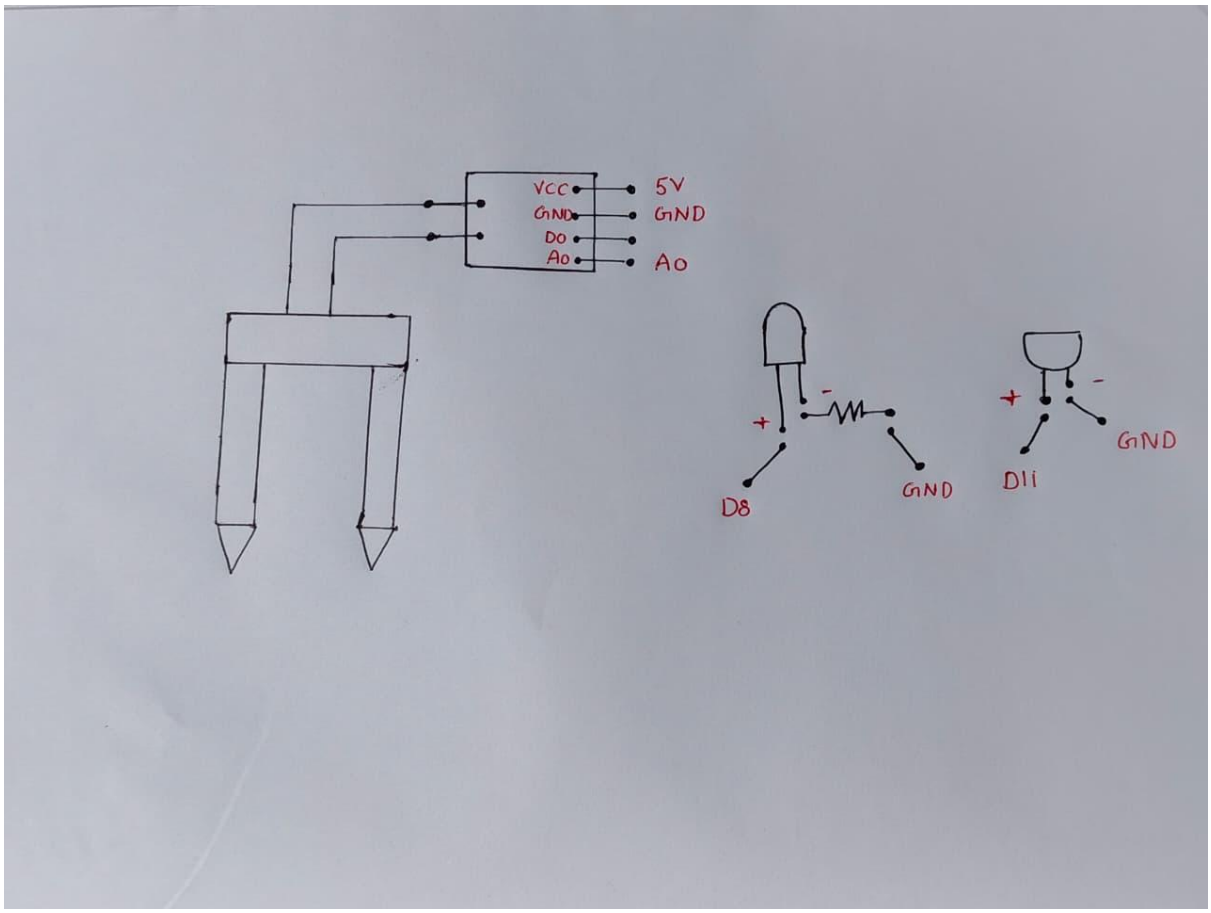
void setup() {
  Serial.begin(9600);
  pinMode(ledPin, OUTPUT);
  pinMode(buzzerPin, OUTPUT);
}

void loop() {
  sensorValue = analogRead(sensorPin);
  Serial.print("Moisture Sensor Value: ");
  Serial.println(sensorValue);

  if (sensorValue < threshold) {
    Serial.println("Status: WET WASTE");
    digitalWrite(ledPin, HIGH);
    digitalWrite(buzzerPin, HIGH);
  } else {
    Serial.println("Status: DRY WASTE");
    digitalWrite(ledPin, LOW);
    digitalWrite(buzzerPin, LOW);
```

```
  }
  delay(1000);
}
```

OUTPUT



7. Develop a program to read the pH value of various substances like milk, lime and water.

```
const int pH_pin = A0;
float voltage, pHValue;

void setup() {
  Serial.begin(9600);
}

void loop() {
  int sensorValue = analogRead(pH_pin);

  // Convert analog value (0–1023) to voltage (0–5V)
  voltage = sensorValue * (5.0 / 1023.0);

  // Convert voltage to pH value
  pHValue = 7 + ((2.5 - voltage) / 0.18);

  Serial.print("Analog: ");
  Serial.print(sensorValue);
  Serial.print(" | Voltage: ");
  Serial.print(voltage, 2);
  Serial.print(" V | pH Value: ");
  Serial.println(pHValue, 2);

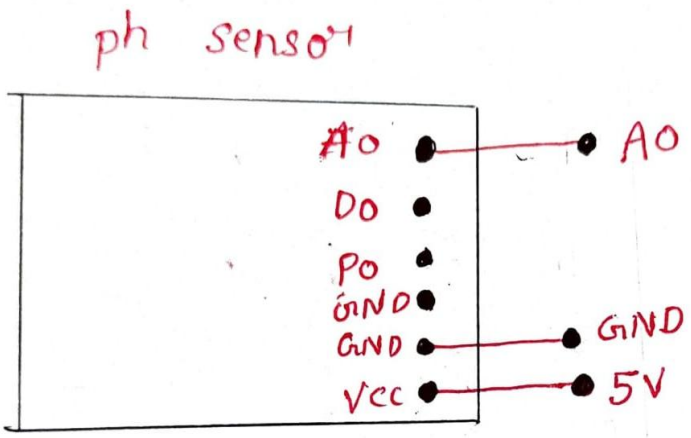
  classifySubstance(pHValue);

  delay(2000);
}

// Function to classify the substance
void classifySubstance(float pH) {
  if (pH < 7.0) {
    Serial.println("Substance is ACIDIC.");
  } else if (pH > 7.0) {
    Serial.println("Substance is ALKALINE.");
  } else {
    Serial.println("Substance is NEUTRAL.");
  }

  Serial.println("---------------------------");
}
```

OUTPUT



8. Develop a program to detect the gas leakage in the surrounding environment.

```
const int gasSensorPin = A0;
const int alarmPin = 8;
int gasLevel = 0;

const int gasThreshold = 400;

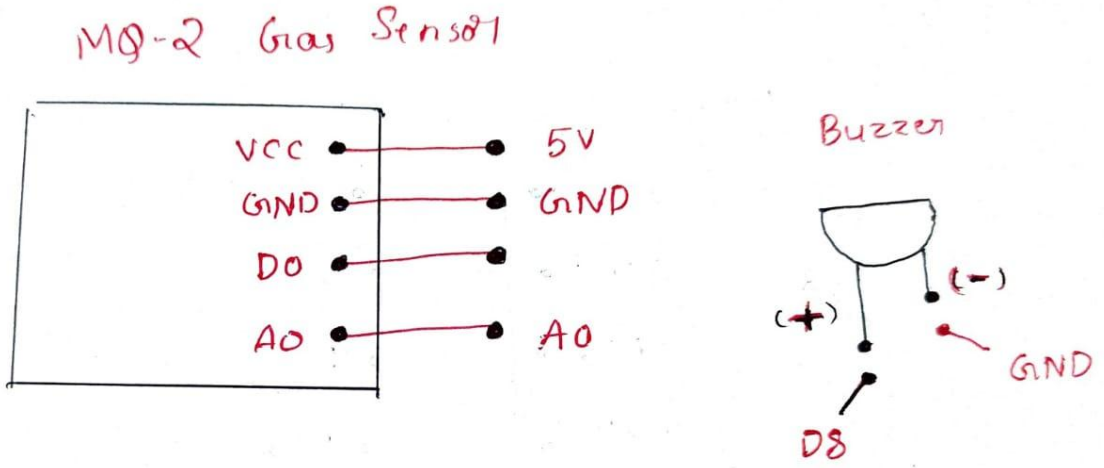
void setup() {
  Serial.begin(9600);
  pinMode(alarmPin, OUTPUT);
}

void loop() {
  gasLevel = analogRead(gasSensorPin);

  Serial.print("Gas Level: ");
  Serial.println(gasLevel);

  if (gasLevel > gasThreshold) {
    digitalWrite(alarmPin, HIGH); // Turn ON alarm
    Serial.println("Gas Leak Detected!");
  } else {
    digitalWrite(alarmPin, LOW);  // Turn OFF alarm
    Serial.println("Safe Environment.");
  }
  delay(1000);
}
```

OUTPUT



9. Develop a program to demonstrate weather station readings using Arduino.

```
#include <Wire.h>
#include <LiquidCrystal_I2C.h>
#include <DHT.h>

#define DHTPIN 2
#define DHTTYPE DHT22

DHT dht(DHTPIN, DHTTYPE);
LiquidCrystal_I2C lcd(0x27, 16, 2);

void setup() {
  lcd.init();
  lcd.backlight();
  dht.begin();
  lcd.setCursor(0,0);
  lcd.print("Weather Station");
  delay(2000);
  lcd.clear();
}

void loop() {
  float temp = dht.readTemperature();
  float hum = dht.readHumidity();

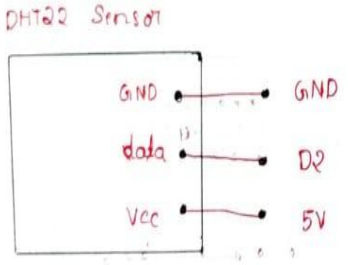
  if (isnan(temp) || isnan(hum)) {
    lcd.setCursor(0, 0);
    lcd.print("Sensor error   ");
    return;
  }

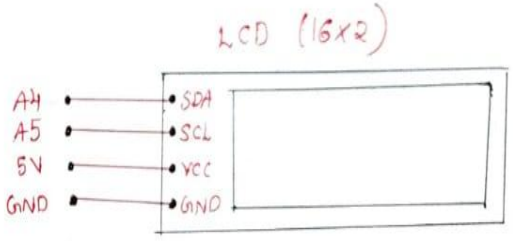
  lcd.setCursor(0, 0);
  lcd.print("Temp: ");
  lcd.print(temp, 1);
  lcd.print((char)223); // Degree symbol
  lcd.print("C");

  lcd.setCursor(0, 1);
  lcd.print("Hum:  ");
  lcd.print(hum, 1);
  lcd.print("%");

  delay(2000); // Wait 2 seconds before next reading
}
```

OUTPUT


DHT22 Sensor
GND
data
Vcc
GND
D2
5V

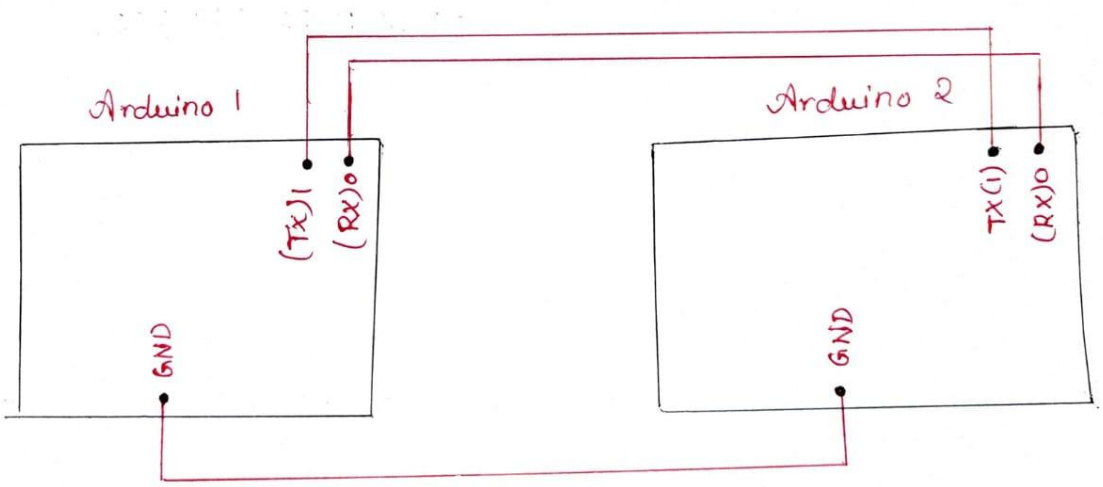

LCD (16x2)
A4
A5
5V
GND
SDA
SCL
Vcc
GND

10. Develop a program to setup a UART protocol and pass a string through the protocol.

```
void setup() {
  Serial.begin(9600);  // Initialize UART at 9600 baud
}

void loop() {
  Serial.println("Hello from Arduino via UART!");
  delay(1000);
}
```

OUTPUT



11. Develop a water level depth detection system using Ultrasonic sensor.

```
const int trigPin = 9;
const int echoPin = 10;

long duration;
float distance_cm;

const float tankHeight_cm = 50.0; // Adjust this to your tank height

void setup() {
  Serial.begin(9600);
  pinMode(trigPin, OUTPUT);
  pinMode(echoPin, INPUT);
}

void loop() {
  // Clear the trigPin by setting it LOW
  digitalWrite(trigPin, LOW);
  delayMicroseconds(2);

  // Send a 10 microsecond pulse to trigger
  digitalWrite(trigPin, HIGH);
  delayMicroseconds(10);
  digitalWrite(trigPin, LOW);

  // Read the echoPin, returns the time in microseconds
  duration = pulseIn(echoPin, HIGH);

  // Calculate distance in cm (speed of sound = 34300 cm/s)
  distance_cm = (duration * 0.0343) / 2;

  // Calculate water level
  float waterLevel = tankHeight_cm - distance_cm;

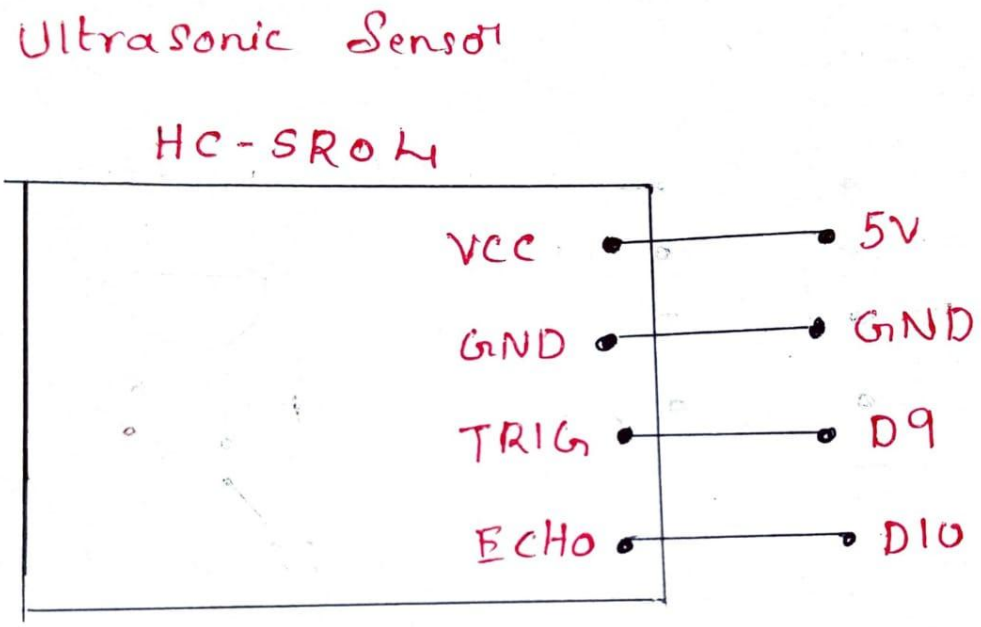
  if (waterLevel < 0) {
    waterLevel = 0; // Water level can't be negative
  }

  Serial.print("Distance to water surface: ");
  Serial.print(distance_cm);
  Serial.println(" cm");

  Serial.print("Water Level: ");
  Serial.print(waterLevel);
  Serial.println(" cm");

  Serial.println("-----------------------");
  delay(1000);
}
```

OUTPUT



12. Develop a program to simulate interfacing with the keypad module to record the keystrokes.

```
#include <Keypad.h>

const byte ROWS = 4; // four rows
const byte COLS = 4; // four columns

// Define the keymap
char keys[ROWS][COLS] = {
  {'1','2','3','A'},
  {'4','5','6','B'},
  {'7','8','9','C'},
  {'*','0','#','D'}
};

// Connect keypad ROW0, ROW1, ROW2 and ROW3 to these Arduino pins.
byte rowPins[ROWS] = {9, 8, 7, 6};

// Connect keypad COL0, COL1, COL2 and COL3 to these Arduino pins.
byte colPins[COLS] = {5, 4, 3, 2};

// Create the Keypad object
Keypad keypad = Keypad( makeKeymap(keys), rowPins, colPins, ROWS, COLS );

String recordedKeys = "";

void setup() {
  Serial.begin(9600);
  Serial.println("Keypad ready. Press keys...");
}

void loop() {
  char key = keypad.getKey();

  if (key) {  // if a key is pressed
    Serial.print("Key Pressed: ");
    Serial.println(key);

    // Append to recorded string
    recordedKeys += key;

    // Print the recorded string
    Serial.print("Recorded keys: ");
    Serial.println(recordedKeys);
  }
}
```

OUTPUT

